

SdCvP-X™: Electronic (Design) Verification Automation Tool

VERIPOINT TECHNOLOGIES PVT LTD

AI Tools & Solutions Powering Semiconductors and Beyond

Company Profile

Veripoint Technologies is a deep-tech product company addressing critical industry challenges, from automating semiconductor verification to delivering scalable, intelligent automation for industrial sectors through its AI-powered tools and engineering solutions.

Industry Challenges

- **Semiconductors:** Manual verification slows development and increases the risk of costly silicon failures.
- **Industrial Sectors:** Scalable, intelligent automation solutions are still limited, hindering operations, efficiency and innovation.

Solutions: Our Flagship Products & Premium DV Services

- **SdCvP-X™** : Proprietary AI-driven Electronic Design Verification Automation Tool.
- **AI-driven solutions** for manufacturing, automotive, telecommunications, healthcare and more.
- **Premium DV services** delivered by a team of experts with deep specialization in verifying complex designs.

Founded: Feb 2022 | **HQ:** Noida, UP, India | **Website:** www.veripointtech.com | **DPIIT:** DIPP158213

SdCvP-X™: Electronic (Design) Verification Automation Tool

SdCvP-X™ is our proprietary AI tool that automates chip design verification, accelerating development and ensuring thorough verification.

Target Designs

- IP (standard & custom), SoC, and Subsystem

Specification (Design)

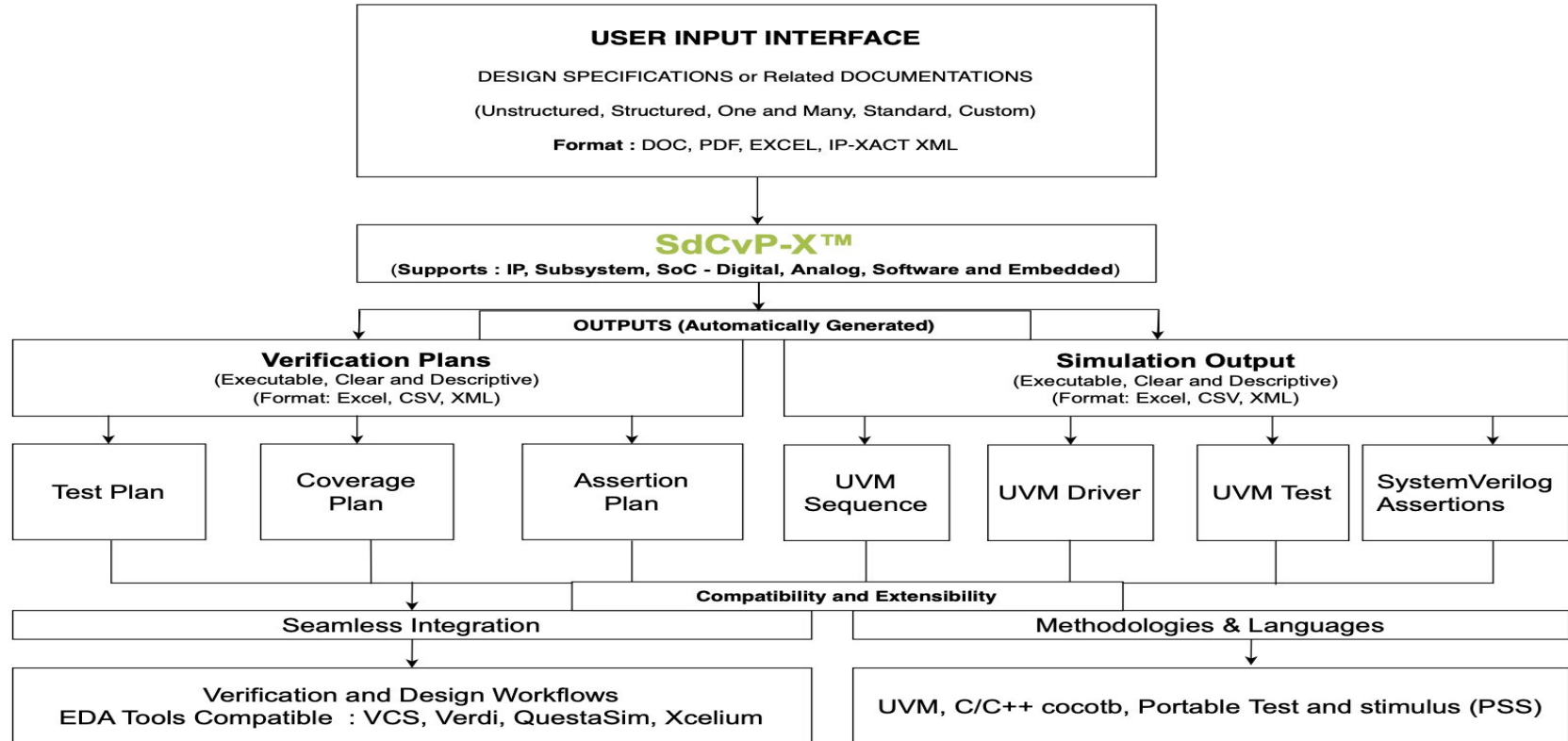
Benchmarks

- High-speed and standard Interfaces like PCIe, CXL, NVMe, Ethernet, UCIe, Display, ARINC
- Custom, Automotive & Networking chip interfaces
- Custom Analog IPs & Systems

Key Features

- **Auto-generates** features, tests, assertions, and coverage plans from specifications in various formats including **IP-XACT XML**
- **Auto-generates** UVM - Sequence, Driver, Test and SystemVerilog Assertions with custom format options
- **Dynamically adapts** to evolving specifications and syncs with manual plans
- **Auto-embeds** debug data for faster root-cause analysis
- **Integrates** with existing verification flows via Vplan for leading EDA tools

SdCvP-X™: Verification Automation Workflow



SdCvP-X™: Verifying Complex Designs

Verification Challenges	SdCvP-X™ Tool
Manual & fragmented planning, tests and assertions gaps	Auto-generates spec-aligned verification plans, UVM - Sequence, Driver, Test and SystemVerilog Assertions for Digital (IP, SoC, Subsystem) and Analog designs, directly from specifications including IP-XACT
Specification Alignment	Plan evolves with specification + Version traceability
Manual Plan Alignment with Specification and upgrades	Aligns manual plans to eliminate gaps and ensure consistency and boost productivity.
Debug Inefficiencies	Traceability matrix + mapping to features/test points
Configuration Explosion	Smart filtering, collapsing, and prioritization
Reuse across derivatives	Scalable planning across standard & custom designs

Unlike EDA simulators (VCS/Xcelium/Quest) that run testbenches, SdCvP-X™ automatically creates spec-aligned executable verification content, reducing manual effort by 50-60% and accelerating development by 30-40%.

Who Benefits: Across the Project Hierarchy

Role	Pain Point	SdCvP-X™ Benefit
DV Engineers	Long debug cycles, manual test list creation	Faster test planning + comprehensive tests and assertions + structured debug : Boosts Productivity
Architects	Hard to validate spec implementation	Full traceability from spec to test
Program/Project Managers	Poor visibility on verification progress	Actionable insights, milestone-based views
Business Leaders & Industries	Risk of re-spin, time-to-market delays	Higher First Silicon Success, faster Time to market

SdCvP-X™: Early Benchmarks Across Target Interfaces

Data Centre & Storage (PCIe, CXL, USB, NVMe, Ethernet IPs & Subsystems)

- Achieved *up to ~60% reduction* in verification planning and execution effort by auto-generating test, coverage, and assertion plans directly from specifications, ensuring spec-alignment.
- Achieved ~50% faster testbench readiness by auto-generating UVM sequences, drivers, and assertions directly from evolving specs.

Automotive & Networking (UCIe, DisplayPort, ARINC, Automotive/Networking IPs, Subsystems & SoCs)

- Achieved ~40% reduction in rework effort by auto-syncing evolving specifications with verification plans, ensuring consistent traceability across derivatives.
- Achieved ~40% faster verification ramp-up by reusing auto-generated verification features across protocol variants.

Mixed-Signal & Custom Systems (Custom Analog IPs & Mixed Systems)

- Achieved ~40% reduction in planning iterations by auto-aligning specifications with tests and assertions for analog and mixed-signal designs.

SdCvP-X™: GUI & OUTPUT for CXL 3.1 IP (Eval) Interface

Design Domain
☒ IP ☐ SoC ☐ Subsystem

Upload Specification
Base Specification (PDF/docs)
CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf
Base Register Specification (.xlsx/PDF)
Base Add-on Specification (PDF/docs)
Base Add-on Register Specification (.xlsx/PDF)

Feature List
Do you want specification feature list extraction?
☒ Yes ☐ No
Download Feature List

Do you want to upload modified feature list according to your design?
☐ Yes ☒ No

Verification Plan Filters
Configuration Enabled
Constraints Enabled
Coverage Enabled
Assertion Enabled
Debug Enabled
UVM - Sequence, Driver, Test Enabled
☒ Yes ☐ No
☒ Yes ☐ No
☒ Yes ☐ No
☒ Yes ☐ No
☒ Yes ☐ No
☒ Yes ☐ No
Confirm Filters
Generate Verification Plan

```

transcript_cxl_3_1.txt
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Domain selected: IP
Uploaded: CXL-3.1-Specification.pdf © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED
Reading Specification: CXL-3.1-Specification © 2019-2023 COMPUTE EXPRESS LINK CONSORTIUM, INC. ALL RIGHTS RESERVED.pdf
Created Processed Data: TEMP_Data_SdCvP-x from Specification CXL-3.1-Specification.pdf
Extracted 16 features.
Feature list saved : /Users/Veripoint_Technologies/Desktop/sdcvp-x/feature_list_cxl_3_1.xlsx
Verification Plan Filters confirmed:
Configuration Enabled: Yes
Constraints Enabled: Yes
Coverage Enabled: Yes
Assertion Enabled: Yes
Debug Enabled: Yes
UVM - Sequence, Driver, Test Enabled: Yes
Verification plan generated and saved: /Users/Veripoint_Technologies/Desktop/sdcvp-x/verification_plan_cxl_3_1.xlsx
Deleted Processed Data: TEMP_Data_SdCvP-x Securing Design Confidentiality
  
```

Verification Plan

Simulation Output

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S.No.	Section	Layer/Component	Feature	DUT Type	Testpoint	Test Scenario	Procedure	Constraint	Assertion	Coverage	Spec Variables	Register Variables
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Verification Plan

Simulation Output

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S.No.	Feature	Test Scenario	UVM Sequence	UVM Driver	UVM Test	SystemVerilog Assertions
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SdCvP-X™: GUI & OUTPUT for Analog PM Bus Interface

Design Domain
☒ IP ☐ SOC ☐ Subsystem

Upload Specification
 Base Specification (PDF/docx)
 Topic_6_Hesse.pdf
 Base Register Specification (.xlsx/PDF)
 Base Add-on Specification (PDF/docx)
 Base Add-on Register Specification (.xlsx/PDF)

Feature List
 Do you want specification feature list extraction? ☒ Yes ☐ No
 Download Feature List

Do you want to upload modified feature list according to your design? ☐ Yes ☒ No

Verification Plan Filters
 Configuration Enabled ☒ Yes ☐ No
 Constraints Enabled ☒ Yes ☐ No
 Coverage Enabled ☒ Yes ☐ No
 Assertion Enabled ☒ Yes ☐ No
 Debug Enabled ☒ Yes ☐ No
 UVM - Sequence, Driver, Test Enabled ☒ Yes ☐ No
 Confirm Filters
 Generate Verification Plan

```
transcript_pm_bus.txt
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Domain selected: IP
Uploaded: Topic_6_Hesse.pdf
Reading Specification: Topic_6_Hesse.pdf
Created Processed Data: TEMP_Data_SdCvP-x from Specification Topic_6_Hesse.pdf
Extracted 33 features.
Feature list saved : /Users/Veripoint_Technologies/Desktop/sdcpv-x/feature_list_pm_bus.xlsx
Verification Plan Filters confirmed:
Configuration Enabled: Yes
Constraints Enabled: Yes
Coverage Enabled: Yes
Assertion Enabled: Yes
Debug Enabled: Yes
UVM - Sequence, Driver, Test Enabled: Yes
Verification plan generated and saved: /Users/Veripoint_Technologies/Desktop/sdcpv-x/verification_plan_pm_bus.xlsx
Deleted Processed Data: TEMP_Data_SdCvP-x Securing Design Confidentiality
```

Verification Plan Simulation Output

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S.No.	Section	Layer/Component	Feature	DUT Type	Testpoint	Test Scenario	Procedure	Constraint	Assertion	Coverage	Spec Variables	Register Variables
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Verification Plan Simulation Output

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S.No.	Feature	Test Scenario	UVM Sequence	UVM Driver	UVM Test	SystemVerilog Assertions
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SdCvP-X™: Tiered Product Options

Standard

- ✓ Comprehensive Verification Plan
- ✓ Features, Test Points & Scenarios
- ✓ Configuration & Constraints Plan

Advanced

- ✓ All Standard Features
- ✓ Assertion Plan
- ✓ Enhanced Specification Evolution
- ✓ Testplan Alignment

Most Popular

Premium

- ✓ All Advanced Features
- ✓ Coverage Plan
- ✓ Auto-Generated UVM - Sequence, Driver, Test and SystemVerilog Assertions
- ✓ Spec Variables, Register Variables, Debug Info & Traceability
- ✓ Priority Support & Success Manager

Contact Us

VERIPOINT TECHNOLOGIES PVT LTD

R&D Office

Unit no. IS-800, Ground Floor, Tower-C, Urbtech Trade Center, Plot No. B-35,
Sector-132, Noida 201304, Uttar Pradesh, India

Sales office

M Floor, WTT Building, C-1, Sector 16, Noida 201301, Uttar Pradesh, India

Tel: +919717580855

Email: rajat@veripointtech.com

Visit: <https://www.veripointtech.com>